

# MICHAL BOREK

(517) 980-3231 • [borekmi1@msu.edu](mailto:borekmi1@msu.edu) • [linkedin.com/in/michal-borek2003/](https://www.linkedin.com/in/michal-borek2003/) • [github.com/michaelborek](https://github.com/michaelborek) Languages: Polish native, English fluent

## AI ENGINEER | MACHINE LEARNING & QUANTITATIVE SYSTEMS

AI Engineer with strong quantitative and software engineering foundations, focused on building reliable AI systems, machine learning pipelines, and automation tools for real-world workflows. Experienced in deep learning, LLM applications, RAG systems, medical imaging ML, HPC tooling, voice AI automation, and client-facing data science projects. Strong in Python, SQL, PyTorch, backend development, model evaluation, validation, and translating ambiguous technical problems into practical AI solutions with measurable impact.

## TECHNICAL SKILLS

---

**Languages:** Python, C++, SQL, Bash

**AI/ML:** PyTorch, scikit-learn, TorchXRayVision, Captum, classification, model evaluation, hyperparameter tuning, calibration

**Quantitative/Data:** Pandas, NumPy, statistical analysis, feature engineering, metrics, F1-score, confusion matrices

**LLM/RAG:** OpenAI Realtime API, Ollama, Qdrant, PostgreSQL Full-Text Search, citation-backed RAG, structured outputs, verification/refusal patterns

**Backend/Systems:** FastAPI, Flask, Docker, Git, Linux, REST APIs, SQLite

**Product/Automation:** Twilio, Electron, React, Cloudflare Workers, ngrok, Nuitka, ShellCheck, Jupyter

**HPC/Distributed:** SLURM, CUDA, PyTorch DDP, Singularity, MSU HPCC

## EDUCATION

---

**Michigan State University, College of Engineering**

Expected May 2026

*Bachelor of Science in Computational Data Science, Minor in Mathematics*

GPA: 3.80

## EXPERIENCE

---

**Institute for Quantitative Health Science & Engineering, Michigan State University**

East Lansing, MI

*Deep Learning Research Assistant, Medical Imaging ML*

Nov 2023 – Present

- Built a modular **PyTorch** pipeline for ordinal medical image classification, supporting reproducible data curation, training, evaluation, and automatic reporting.
- Led hyperparameter optimization and model selection using held-out metrics, calibration analysis, and structured performance tracking.
- Scaled model training with **PyTorch DDP** on CUDA/SLURM, reducing wall time from approximately **8 hours to 3 hours**.
- Improved preprocessing efficiency through optimized image resizing and pipeline design for large-scale chest X-ray experiments.
- Supported clinical model review with **Captum**-based saliency analysis to improve interpretability and explainability.
- Contributed to research disseminated through **SPIE Medical Imaging 2025** and a **Radiology: Artificial Intelligence** manuscript currently under review.

**Institute for Cyber-Enabled Research, Michigan State University**

East Lansing, MI

*AI Engineer Intern, HPC / ML Tooling*

Jun 2023 – May 2026

- Shipped an AI-assisted **SLURM job-script validator** combining custom #SBATCH checks, ShellCheck, and local LLM review through Ollama to reduce submission errors and speed user triage.
- Deployed a **Flask service and CLI** on Open OnDemand with a live script editor, real-time analysis, and SLURM-focused chat assistant for the HPCC user community.
- Designed **policy-constrained LLM workflows** with approved checks, controlled outputs, and validation logic to improve reliability and reduce unsupported guidance.
- Authored an HPCC module for **LM Studio**, enabling one-command LLM workflows under SLURM and supporting local inference experiments on H200 GPUs.

- Presented the AI agent at **Mid-SURE 2025** through a live demo and Q&A with faculty, students, and technical stakeholders.
- Designed and delivered a **Python training track** for HPC users covering programming fundamentals, OOP, NumPy, and project structure.

PROJECTS

---

*AI Auto-Caller for Outbound Prospecting* Mar 2026 – Present

- Built a local macOS desktop app using **Electron, React, FastAPI, and SQLite** to automate outbound sales calls with OpenAI Realtime API and Twilio.
- Shipped the product as a one-click **.dmg** with bundled ngrok, allowing non-technical customers to run the system with zero infrastructure setup.
- Designed a **licensing and anti-piracy system** using a Cloudflare Worker kill-switch with HMAC-signed responses.
- Validated the system against a **Level-2 pentest** and closed all 8 identified attack vectors.
- Built a production-grade release pipeline with **137 passing Python tests**, automated release scripts, and a Nuitka-compiled engine binary of approximately 60 MB.

*Legal Document Intelligence System* Dec 2025 – Feb 2026

- Built an **end-to-end RAG** system for PDF, DOCX, and TXT legal documents with citation-backed answers using **PostgreSQL Full-Text Search, Qdrant**, and local Ollama inference.
- Implemented **structured outputs, verification/refusal logic**, claim-to-source validation, excerpt generation, audit logging, retries, and safe concurrency.

*Delta Dental Capstone – Project Lead (Client-Facing Data Science)* Jan 2026 – Apr 2026

- Led a partner-sponsored data science capstone with the **Delta Dental Data Science Team**, coordinating scope, milestones, working sessions, and stakeholder updates.
- Built a **canonical dataset** by consolidating source tables into a stable schema and data dictionary for downstream analytics.
- Prototyped **feature mapping into a 3D model** to improve interpretability, visualization, and feature semantics.

PUBLICATIONS

---

- **Liu, M.**, Loveless, I., Huang, Z., **Borek, M.**, Rosenman, K., Alessio, A., Wang, L. “Ordinal classification framework for multiclass grading of pneumoconiosis.” *SPIE Medical Imaging 2025: Computer-Aided Diagnosis*, 13407:134072Q, April 2025. **doi:10.1117/12.3046353**
- **Wang, L.**, Liu, M., Huang, Z., Loveless, I., **Borek, M.**, Rosenman, K., Alessio, A. “Pneumoconiosis Multi-task Screening and Classification using Fine-Tuned Deep Learning Models.” *Radiology: Artificial Intelligence*, under review, 2025.

LEADERSHIP & AWARDS

---

**MSU Housing & Residence Education** East Lansing, MI  
*Resident Assistant* Aug 2024 – May 2026

- Supported **100+ residents** through mentoring, conflict mediation, incident response, and community programming; recognized with **REHS Excellent Teamwork Award** in 2025.

**Polish Club, Michigan State University** East Lansing, MI  
*Co-Founder / Executive Board Member* Oct 2024 – May 2026

- Co-founded a student organization supporting Polish culture, student community, campus engagement, outreach, and event planning.

**Awards:** REHS Excellent Teamwork Award, 2025 • Dean’s List, 7/8 terms • EGRID Silver Scholarship, 2024 • International Tuition Grant, 2022